

Hamiltonian of Bilayer Graphene

October 14, 2020

1 Tight-binding model

The geometry of a bilayer graphene with AB stacking is shown in Fig.1. There are four atoms in each unit cell: A1, B1, A2, and B2, where 1 and 2 refers to the top and bottom layer and A and B refers to the sublattice indices. The primitive lattice vectors are

$$\mathbf{a}_1 = \left(\frac{a}{2}, \frac{\sqrt{3}a}{2} \right), \quad \mathbf{a}_2 = \left(\frac{a}{2}, -\frac{\sqrt{3}a}{2} \right) \quad (1)$$

and the vectors connecting the nearest neighbours in each layer are

$$\boldsymbol{\delta}_1 = \left(0, \frac{a}{\sqrt{3}} \right), \quad \boldsymbol{\delta}_2 = \left(\frac{a}{2}, -\frac{a}{2\sqrt{3}} \right), \quad \boldsymbol{\delta}_3 = \left(-\frac{a}{2}, -\frac{a}{2\sqrt{3}} \right) \quad (2)$$

In the tight binding model, the basis function for each site can be written as

$$\chi_{\mathbf{k},\alpha}(\mathbf{r}) = \frac{1}{N} \sum_i e^{i\mathbf{k} \cdot \mathbf{R}_{i\alpha}} \phi(\mathbf{r} - \mathbf{R}_{i\alpha}), \quad (3)$$

where $\alpha = \{A1, B1, A2, B2\}$ sites, 1 refers to the bottom layer, 2 refers to the top layer, i is the index for the unit cell, $\mathbf{R}_{i\alpha}$ is the position for each site and $\phi(\mathbf{r} - \mathbf{R}_{i\alpha})$ is the atomic orbital. The wavefunction is given by

$$\psi_{\mathbf{k}} = \sum_{\alpha} c_{\alpha} \chi_{\mathbf{k},\alpha}. \quad (4)$$

The band structure is given by

$$\begin{aligned} \langle \chi_{\mathbf{k},\alpha} | H | \psi_{\mathbf{k}} \rangle &= \frac{1}{N} \sum_{i,j} c_{A1} e^{i\mathbf{k} \cdot (\mathbf{R}_{i,A1} - \mathbf{R}_{j,\alpha})} \langle \phi(\mathbf{r} - \mathbf{R}_{j,\alpha}) | H | \phi(\mathbf{r} - \mathbf{R}_{i,A1}) \rangle + c_{B1} e^{i\mathbf{k} \cdot (\mathbf{R}_{i,B1} - \mathbf{R}_{j,\alpha})} \langle \phi(\mathbf{r} - \mathbf{R}_{j,\alpha}) | H | \phi(\mathbf{r} - \mathbf{R}_{i,B1}) \rangle \\ &+ c_{A2} e^{i\mathbf{k} \cdot (\mathbf{R}_{i,A2} - \mathbf{R}_{j,\alpha})} \langle \phi(\mathbf{r} - \mathbf{R}_{j,\alpha}) | H | \phi(\mathbf{r} - \mathbf{R}_{i,A2}) \rangle + c_{B2} e^{i\mathbf{k} \cdot (\mathbf{R}_{i,B2} - \mathbf{R}_{j,\alpha})} \langle \phi(\mathbf{r} - \mathbf{R}_{j,\alpha}) | H | \phi(\mathbf{r} - \mathbf{R}_{i,B2}) \rangle \\ &= E_{\mathbf{k}} \langle \chi_{\mathbf{k},\alpha} | \psi_{\mathbf{k}} \rangle. \end{aligned} \quad (5)$$

We assume that the overlap integrals between different sites are zero: $\langle \phi(\mathbf{r} - \mathbf{R}_{i\alpha}) | \phi(\mathbf{r} - \mathbf{R}_{i'\alpha'}) \rangle = \delta_{i,i'} \delta_{\alpha,\alpha'}$.

The onsite energy is given by $H_{\alpha\alpha} = \frac{1}{N} \sum_i \langle \phi(\mathbf{r} - \mathbf{R}_{i,\alpha}) | H | \phi(\mathbf{r} - \mathbf{R}_{i,\alpha}) \rangle$. For simplicity, we set $H_{A1,A1} = H_{B1,B1} = \Delta/2$ and $H_{A2,A2} = H_{B2,B2} = \Delta/2$, where Δ is the potential difference between two layers. Furthermore, we only consider nearest-neighbour hopping. The four-band Hamiltonian is thus given by

$$H^{AB} = \begin{pmatrix} \frac{\Delta}{2} & -\gamma_0 f(\mathbf{k}) & \gamma_4 f(\mathbf{k}) & -\gamma_3 f^*(\mathbf{k}) \\ -\gamma_0 f^*(\mathbf{k}) & \frac{\Delta}{2} & \gamma_1 & \gamma_4 f(\mathbf{k}) \\ \gamma_4 f^*(\mathbf{k}) & \gamma_1 & -\frac{\Delta}{2} & -\gamma_0 f(\mathbf{k}) \\ -\gamma_3 f(\mathbf{k}) & \gamma_4 f^*(\mathbf{k}) & -\gamma_0 f^*(\mathbf{k}) & -\frac{\Delta}{2} \end{pmatrix} \quad (6)$$

where

$$\begin{aligned} f(\mathbf{k}) &= \sum_{l=1}^3 e^{i\mathbf{k} \cdot \boldsymbol{\delta}_l} = e^{i\frac{a}{\sqrt{3}}k_y} + e^{i(\frac{a}{2}k_x - \frac{a}{2\sqrt{3}}k_y)} + e^{i(-\frac{a}{2}k_x - \frac{a}{2\sqrt{3}}k_y)} \\ &= e^{i\frac{a}{\sqrt{3}}k_y} + 2e^{-i\frac{a}{2\sqrt{3}}k_y} \cos\left(\frac{a}{2}k_x\right). \end{aligned} \quad (7)$$

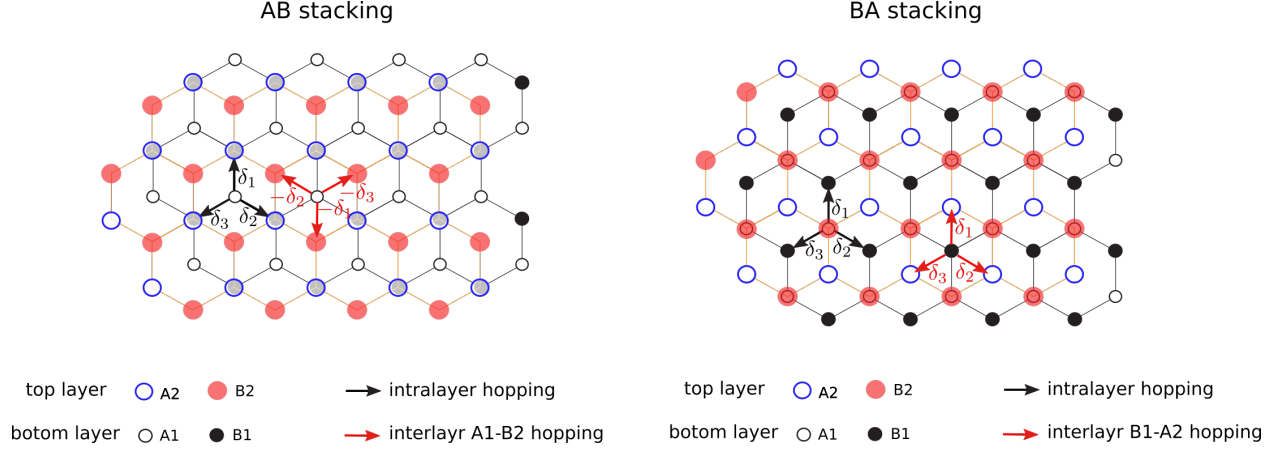


Figure 1: Geometry of bilayer graphene with AB stacking

Note that the interlayer non-dimer interaction term $H_{A1,B2}$ involves hopping from the A1 site to the B2 site with hopping vector $-\delta_1$, $-\delta_2$ and $-\delta_3$, thus $H_{A1,B2}$ contains $\sum_{l=1}^3 e^{i\mathbf{k}\cdot(-\delta_l)} = f^*(\mathbf{k})$.

For BA stacking, one of the layer is shifted by half a unit cell and we have B2 atom directly on top of A1. The non-dimer interaction term $H_{B1,A2}$ now involves hopping between the B1 site to the A2 site given by the vectors δ_1 , δ_2 and δ_3 , thus the Hamiltonian is

$$H^{BA} = \begin{pmatrix} \frac{\Delta}{2} & -\gamma_0 f(\mathbf{k}) & \gamma_4 f^*(\mathbf{k}) & \gamma_1 \\ -\gamma_0 f^*(\mathbf{k}) & \frac{\Delta}{2} & -\gamma_3 f(\mathbf{k}) & \gamma_4 f^*(\mathbf{k}) \\ \gamma_4 f(\mathbf{k}) & -\gamma_3 f^*(\mathbf{k}) & -\frac{\Delta}{2} & -\gamma_0 f(\mathbf{k}) \\ \gamma_1 & \gamma_4 f(\mathbf{k}) & -\gamma_0 f^*(\mathbf{k}) & -\frac{\Delta}{2} \end{pmatrix} \quad (8)$$

For $\Delta = 0$, the energy gap is minimal at K and K' points. The K and K' points are determined by setting $f(\mathbf{k}) = 0$, which gives

$$\cos\left(\frac{a}{2}k_x\right) = -\frac{1}{2}e^{i\frac{\sqrt{3}a}{2}k_y} = -\frac{1}{2}\left[\cos\left(\frac{\sqrt{3}a}{2}k_x + i\sin\left(\frac{\sqrt{3}a}{2}k_x\right)\right)\right] \quad (9)$$

$$\Rightarrow \frac{\sqrt{3}a}{2}k_y = 0, \pm\pi \quad (10)$$

For $k_y = 0$, we have $k_x = \pm\frac{4\pi}{3a}$. For $k_y = \pm\frac{2\pi}{\sqrt{3}a}$, we have $k_x = \pm\frac{2\pi}{3a}$.

Now we choose to expand the Hamiltonian around $\mathbf{K}_{\pm} = (\pm\frac{4\pi}{3a}, 0)$.

$$\frac{df}{dk_x} = -ae^{-i\frac{a}{2\sqrt{3}}k_y} \sin\left(\frac{a}{2}k_x\right) \quad (11)$$

$$\frac{df}{dk_y} = i\frac{a}{\sqrt{3}}e^{i\frac{a}{\sqrt{3}}k_y} - \frac{ia}{\sqrt{3}}e^{-i\frac{a}{2\sqrt{3}}k_y} \cos\left(\frac{a}{2}k_x\right) \quad (12)$$

At the $\mathbf{K}_{\pm}$ points,

$$\left.\frac{df}{d\mathbf{k}}\right|_{\mathbf{K}_{\pm}} = \frac{\sqrt{3}a}{2}(\mp 1, i) \quad (13)$$

Thus up to linear order, we can right

$$f(\mathbf{k}) = \mathbf{p} \cdot \left.\frac{df}{d\mathbf{k}}\right|_{\mathbf{K}_{\pm}} = -\frac{\sqrt{3}a}{2}(\pm p_x - ip_y) \quad (14)$$

(If we choose to expand the Hamiltonian at $\mathbf{K}_\pm = \left(\mp \frac{2\pi}{3a}, \frac{2\pi}{\sqrt{3}a}\right)$, then

$$\left. \frac{df}{d\mathbf{k}} \right|_{\mathbf{K}_\pm} = \frac{\sqrt{3}a}{2} e^{i\pi/3} (\pm 1, -i), \quad (15)$$

and

$$f(\mathbf{k}) = \mathbf{p} \cdot \left. \frac{df}{d\mathbf{k}} \right|_{\mathbf{K}_\pm} = \frac{\sqrt{3}a}{2} (\pm p_x - ip_y) e^{i\pi/3} \quad (16)$$

There is a phase in the expansion of $f(\mathbf{k})$.

Now we expand $f(\mathbf{k})$ to the second order to obtain the quadratic trigonal warping term.

$$\frac{d^2 f}{dk_x^2} = -\frac{a^2}{2} e^{-i\frac{k_y a}{2\sqrt{3}}} \cos\left(\frac{k_x a}{2}\right) \quad (17)$$

$$\frac{d^2 f}{dk_x^2} = -\frac{a^2}{3} e^{i\frac{k_y a}{\sqrt{3}}} - \frac{a^2}{6} e^{-i\frac{k_y a}{2\sqrt{3}}} \cos\left(\frac{k_x a}{2}\right) \quad (18)$$

$$\frac{d^2 f}{dk_x k_y} = i \frac{a^2}{2\sqrt{3}} e^{-i\frac{k_y a}{2\sqrt{3}}} \sin\left(\frac{k_x a}{2}\right) \quad (19)$$

At $\mathbf{K}_\pm = (\pm \frac{4\pi}{3a}, 0)$, we have

$$\frac{d^2 f}{dk_x^2} = \frac{a^2}{4}, \quad -\frac{d^2 f}{dk_x^2} = \frac{a^2}{4}, \quad \frac{d^2 f}{dk_x k_y} = \pm i \frac{a^2}{4} \quad (20)$$

Thus the quadratic term in the expansion of $f(\mathbf{k})$ is

$$\frac{a^2}{4} (k_x^2 - k_y^2 \pm 2ik_x k_y) \quad (21)$$

To the first order of p , the four-band Hamiltonian in the basis of $\{\psi_{A1}, \psi_{B1}, \psi_{A2}, \psi_{B2}\}$ and setting $\gamma_4 = 0$ is

$$H^{AB} = \begin{pmatrix} \frac{\Delta}{2} & v\pi^\dagger & 0 & v_3\pi \\ v\pi & \frac{\Delta}{2} & \gamma_1 & 0 \\ 0 & \gamma_1 & -\frac{\Delta}{2} & v\pi^\dagger \\ v_3\pi^\dagger & 0 & v\pi & -\frac{\Delta}{2} \end{pmatrix}, \quad H^{BA} = \begin{pmatrix} \frac{\Delta}{2} & v\pi^\dagger & 0 & \gamma_1 \\ v\pi & \frac{\Delta}{2} & v_3\pi^\dagger & 0 \\ 0 & v_3\pi & -\frac{\Delta}{2} & v\pi^\dagger \\ \gamma_1 & 0 & v\pi & -\frac{\Delta}{2} \end{pmatrix}, \quad (22)$$

where $\pi = \xi p_x + ip_y$.

2 Strain

The strain modifies the hopping constant along different hopping directions. For a uniaxial strain in the direction $\mathbf{l} = (\sin 2\theta, \cos 2\theta)$, the modified intralayer A-B sublattice hopping along the direction δ_l is given by

$$\tilde{\gamma}_0^l = \left[1 + \left(\frac{\delta' - \delta}{2} \frac{\delta_l}{a_0} \cdot \mathbf{l} + \frac{\delta + \delta'}{2} \right) \eta_0 \right] \gamma_0, \quad (23)$$

where $a_0 = a/\sqrt{3}$ is the carbon-carbon bond length. Thus the intralayer A-B coupling is

$$\begin{aligned} H_{A1,B1}^{(AB)} &= \sum_{l=1}^3 -\tilde{\gamma}_0^l e^{i\mathbf{k} \cdot \delta_l} \\ &= -\left(1 + \frac{\delta + \delta'}{2} \eta_0 \right) \gamma_0 \sum_{l=1}^3 (e^{i\mathbf{k} \cdot \delta_l}) - \frac{\delta' - \delta}{2} \eta_0 \gamma_0 \sum_{l=1}^3 \frac{\sqrt{3}\delta_l}{a} \cdot \mathbf{l} e^{i\mathbf{k} \cdot \delta_l} \\ &= -\left(1 + \frac{\delta + \delta'}{2} \eta_0 \right) \gamma_0 f(\mathbf{k}) \\ &\quad + \frac{\delta - \delta'}{2} \eta_0 \gamma_0 \left[\cos 2\theta e^{i\frac{a}{\sqrt{3}}k_y} + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(\frac{a}{2}k_x - \frac{a}{2\sqrt{3}}k_y)} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(-\frac{a}{2}k_x - \frac{a}{2\sqrt{3}}k_y)} \right] \end{aligned} \quad (24)$$

We now expand $H_{A1,B1}^{(AB)}$ in the vicinity of $\mathbf{K}_\xi = (\xi \frac{4\pi}{3a}, 0)$ points. We are interested in the correction due to strain independent of $\mathbf{p} = \mathbf{k} - \mathbf{K}_\xi$, thus we substitute $\mathbf{k} = (\xi \frac{4\pi}{3a}, 0)$ to the term in the square bracket:

$$\begin{aligned}
H_{A1,B1}^{(AB)} &= \gamma_0 \pi^\dagger + \frac{\delta - \delta'}{2} \eta_0 \gamma_0 \left[\cos 2\theta + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i\frac{a}{2}k_x} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{-i\frac{a}{2}k_x} \right] \\
&= \gamma_0 \pi^\dagger + \frac{\delta - \delta'}{2} \eta_0 \gamma_0 \left[\cos 2\theta + i\sqrt{3} \sin 2\theta \sin \left(\frac{a}{2}k_x \right) - \cos 2\theta \cos \left(\frac{a}{2}k_x \right) \right] \\
&= \gamma_0 \pi^\dagger + \frac{\delta - \delta'}{2} \eta_0 \gamma_0 \left[\cos 2\theta + \xi \frac{3}{2} i \sin 2\theta + \frac{1}{2} \cos 2\theta \right] \\
&= \gamma_0 \pi^\dagger + \frac{3}{4} (\delta - \delta') \eta_0 \gamma_0 \theta e^{i\xi 2\theta}
\end{aligned} \tag{25}$$

This agrees with Eq. (33).

According to Fal'ko, the $A1 - B2$ hopping is given by

$$\tilde{\gamma}_3^l = \left[1 + \left(\frac{\delta' - \delta}{2} \frac{\boldsymbol{\delta}_l}{a_0} \cdot \mathbf{1} - \frac{\boldsymbol{\delta}_l}{a_0} \cdot \frac{\delta \mathbf{r}}{a_0} + \frac{\delta + \delta'}{2} \right) \eta_3 \right] \gamma_3, \tag{26}$$

where $\delta \mathbf{r} = \rho a (\sin \varphi, -\cos \varphi)$ is the interlayer shear and $\rho = \delta r / a$. Thus the interlayer coupling is

$$\begin{aligned}
H_{A1,B2}^{(AB)} &= \sum_{l=1}^3 -\tilde{\gamma}_3^l e^{i\mathbf{k} \cdot (-\boldsymbol{\delta}_l)} \\
&= - \left(1 + \frac{\delta + \delta'}{2} \eta_3 \right) \gamma_3 \sum_{l=1}^3 (e^{-i\mathbf{k} \cdot \boldsymbol{\delta}_l}) - \frac{\delta' - \delta}{2} \eta_3 \gamma_3 \sum_{l=1}^3 \frac{\sqrt{3} \boldsymbol{\delta}_l}{a} \cdot \mathbf{1} e^{-i\mathbf{k} \cdot \boldsymbol{\delta}_l} + 3 \eta_3 \gamma_3 \sum_{l=1}^3 \frac{\boldsymbol{\delta}_l}{a} \cdot \frac{\delta \mathbf{r}}{a} e^{-i\mathbf{k} \cdot \boldsymbol{\delta}_l} \\
&= - \left(1 + \frac{\delta + \delta'}{2} \eta_3 \right) \gamma_3 f(\mathbf{k}) \\
&+ \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta e^{-i\frac{a}{\sqrt{3}}k_y} + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(-\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} \right] \\
&+ \sqrt{3} \rho \eta_3 \gamma_3 \left[-\cos \varphi e^{-i\frac{a}{\sqrt{3}}k_y} + \left(\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i(-\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} + \left(-\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i(\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} \right]
\end{aligned} \tag{27}$$

Again, we expand the Hamiltonian near $\mathbf{K}_\xi$. For the correction due to strain, we are only concerned with the term independent of $\mathbf{p}$:

$$\begin{aligned}
H_{A1,B2}^{(AB)} &= \gamma_3 \pi + \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{-i\frac{a}{2}k_x} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i\frac{a}{2}k_x} \right] \\
&+ \sqrt{3} \rho \eta_3 \gamma_3 \left[-\cos \varphi + \left(\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{-i\frac{a}{2}k_x} + \left(-\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i\frac{a}{2}k_x} \right] \\
&= \gamma_3 \pi + \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta - i\xi \sqrt{3} \sin 2\theta \sin \left(\frac{2\pi}{3} \right) - \cos 2\theta \cos \left(\frac{2\pi}{3} \right) \right] \\
&+ \sqrt{3} \rho \eta_3 \gamma_3 \left[-\cos \varphi - i\xi \sqrt{3} \sin \varphi \sin \left(\frac{2\pi}{3} \right) + \cos \varphi \cos \left(\frac{2\pi}{3} \right) \right] \\
&= \gamma_3 \pi + \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta - i\xi \frac{3}{2} \sin 2\theta + \frac{1}{2} \cos 2\theta \right] + \sqrt{3} \rho \eta_3 \gamma_3 \left[-\cos \varphi - i\xi \frac{3}{2} \sin \varphi - \frac{1}{2} \cos \varphi \right] \\
&= \gamma_3 \pi + \frac{3}{4} (\delta - \delta') \eta_3 \gamma_3 e^{-i\xi 2\theta} - \frac{3\sqrt{3}}{2} \rho \eta_3 \gamma_3 e^{i\xi \varphi}
\end{aligned} \tag{28}$$

This agrees with Eq. (34).

—————my thoughts on the modified $\tilde{\gamma}_3$ —————

However, the hopping vectors for A1-B2 hopping are $-\delta_l$, thus my postulate for the modified hopping paramter is

$$\tilde{\gamma}_3^l = \left[1 + \left(\frac{\delta' - \delta}{2} \frac{(-\delta_l)}{a_0} \cdot \mathbf{1} - \frac{(-\delta_l)}{a_0} \cdot \frac{\delta \mathbf{r}}{a_0} + \frac{\delta + \delta'}{2} \right) \eta_3 \right] \gamma_3. \quad (29)$$

The Hamiltonian matrix element is thus

$$\begin{aligned} H_{A1,B2}^{(AB)} &= \sum_{l=1}^3 -\tilde{\gamma}_3^l e^{i\mathbf{k} \cdot (-\delta_l)} \\ &= - \left(1 + \frac{\delta + \delta'}{2} \eta_3 \right) \gamma_3 \sum_{l=1}^3 (e^{-i\mathbf{k} \cdot \delta_l}) + \frac{\delta' - \delta}{2} \eta_3 \gamma_3 \sum_{l=1}^3 \frac{\sqrt{3}\delta_l}{a} \cdot \mathbf{1} e^{-i\mathbf{k} \cdot \delta_l} - 3\eta_3 \gamma_3 \sum_{l=1}^3 \frac{\delta_l}{a} \cdot \frac{\delta \mathbf{r}}{a} e^{-i\mathbf{k} \cdot \delta_l} \\ &= - \left(1 + \frac{\delta + \delta'}{2} \eta_3 \right) \gamma_3 f(\mathbf{k}) \\ &\quad - \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta e^{-i\frac{a}{\sqrt{3}}k_y} + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(-\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i(\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} \right] \\ &\quad - \sqrt{3}\rho\eta_3\gamma_3 \left[-\cos \varphi e^{-i\frac{a}{\sqrt{3}}k_y} + \left(\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i(-\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} + \left(-\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i(\frac{a}{2}k_x + \frac{a}{2\sqrt{3}}k_y)} \right] \end{aligned} \quad (30)$$

Near the $\mathbf{K}_\xi$ points, we have

$$\begin{aligned} H_{A1,B2}^{(AB)} &= \gamma_3 \pi - \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta + \left(\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{-i\frac{a}{2}k_x} + \left(-\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right) e^{i\frac{a}{2}k_x} \right] \\ &\quad - \sqrt{3}\rho\eta_3\gamma_3 \left[-\cos \varphi + \left(\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{-i\frac{a}{2}k_x} + \left(-\frac{\sqrt{3}}{2} \sin \varphi + \frac{1}{2} \cos \varphi \right) e^{i\frac{a}{2}k_x} \right] \\ &= \gamma_3 \pi - \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta - i\xi\sqrt{3} \sin 2\theta \sin \left(\frac{2\pi}{3} \right) - \cos 2\theta \cos \left(\frac{2\pi}{3} \right) \right] \\ &\quad - \sqrt{3}\rho\eta_3\gamma_3 \left[-\cos \varphi - i\xi\sqrt{3} \sin \varphi \sin \left(\frac{2\pi}{3} \right) + \cos \varphi \cos \left(\frac{2\pi}{3} \right) \right] \\ &= \gamma_3 \pi - \frac{\delta - \delta'}{2} \eta_3 \gamma_3 \left[\cos 2\theta - i\xi\frac{3}{2} \sin 2\theta + \frac{1}{2} \cos 2\theta \right] - \sqrt{3}\rho\eta_3\gamma_3 \left[-\cos \varphi - i\xi\frac{3}{2} \sin \varphi - \frac{1}{2} \cos \varphi \right] \\ &= \gamma_3 \pi - \frac{3}{4}(\delta - \delta')\eta_3\gamma_3 e^{-i\xi 2\theta} + \frac{3\sqrt{3}}{2}\rho\eta_3\gamma_3 e^{i\xi\varphi} \end{aligned} \quad (31)$$

—————end of my guess—————

—————From Fal'ko—————

The Hamiltonian for strained BLG in AB stacking can be written as

$$\begin{aligned}
H^{AB} &= \begin{pmatrix} \frac{\Delta}{2} & v\pi^\dagger + A_0^* & 0 & v_3\pi + A_3 \\ v\pi + A_0 & \frac{\Delta}{2} & \gamma_1 & 0 \\ 0 & \gamma_1 & -\frac{\Delta}{2} & v\pi^\dagger + A_0^* \\ v_3\pi^\dagger + A_3^* & 0 & v\pi + A_0 & -\frac{\Delta}{2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{\Delta}{2} & v(\pi + \frac{A_0}{v}) & v(\pi^\dagger + \frac{A_0^*}{v}) & 0 & v_3(\pi + \frac{A_0}{v}) + A_3 - \frac{v_3 A_0}{v} \\ v(\pi + \frac{A_0}{v}) & \frac{\Delta}{2} & \gamma_1 & 0 & 0 \\ 0 & \gamma_1 & -\frac{\Delta}{2} & v(\pi^\dagger + \frac{A_0^*}{v}) & 0 \\ v_3(\pi^\dagger + \frac{A_0^*}{v}) + A_3^* - \frac{v_3 A_0^*}{v} & 0 & v(\pi + \frac{A_0}{v}) & -\frac{\Delta}{2} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \frac{\Delta}{2} & v\tilde{\pi}^\dagger & 0 & v_3\tilde{\pi} + A \\ v\tilde{\pi} & \frac{\Delta}{2} & \gamma_1 & 0 \\ 0 & \gamma_1 & -\frac{\Delta}{2} & v\tilde{\pi}^\dagger \\ v_3\tilde{\pi}^\dagger + A^* & 0 & v\tilde{\pi} & -\frac{\Delta}{2} \end{pmatrix}, \tag{32}
\end{aligned}$$

where

$$A_0 = \frac{3}{4}e^{-i2\xi\theta}(\delta - \delta')\eta_0\gamma_0 \tag{33}$$

and

$$A_3 = \frac{3}{4}e^{-i2\xi\theta}(\delta - \delta')\eta_3\gamma_0 - \frac{3\sqrt{3}}{2}\frac{\delta r}{a}e^{i\xi\varphi}\eta_3\gamma_3 \tag{34}$$

Thus, the contribution of strain to the skew interlayer coupling is given by

$$A = \frac{3}{4} \left[e^{-i2\xi\theta}(\delta - \delta')(\eta_3 - \eta_0) - 2\sqrt{3}e^{i\xi\varphi}\rho\eta_3 \right] \gamma_3, \tag{35}$$

where the first term in the square bracket is induced by straining of the lattice, while the second term is induced by interlayer shear (relative shift of the two layers) associated with the strain. We consider a uniaxial strain with its principle axis with an angle θ with respect to the x (zigzag) direction. Here, δ and $\delta' = -0.165\delta$ correspond to the extension along the principle axis and the contraction in the direction perpendicular to the principle axis. $\eta_0 \sim -3$ and $\eta_3 \sim -1$ are the Grüneisen parameters. For uniaxial strain, the strength and direction of the shear deformation are given by $\rho = \delta r/a = 0.336\delta\frac{\eta_0 - \eta_3}{\eta_3}$ and $\varphi = \pi - 2\theta$.

Thus, we can simplify A :

$$\begin{aligned}
A &= \frac{3}{4} \left[2e^{-i2\xi\theta}(\delta - \delta') + 2\sqrt{3}e^{-i2\xi\theta}e^{i\pi}\rho \right] \gamma_3 \\
&= \frac{3}{4} \left[2(\delta - \delta') - 2\sqrt{3}\rho \right] \gamma_3 e^{-i2\xi\theta} \\
&= |A|e^{-i2\xi\theta} = |A|e^{i\xi(\varphi - \pi)} \tag{36}
\end{aligned}$$

3 Two-band approximation

We derive the two-band Hamiltonian in the basis of non-dimer wavefunctions.

3.1 AB stacking

In AB stacking, the non-dimer basis is $\theta = \{\psi_{A1}, \psi_{B2}\}$, and the dimer basis is $\chi = \{\psi_{A2}, \psi_{B1}\}$. We rewrite the Hamiltonian in the basis of $\{\psi_{A1}, \psi_{B2}, \psi_{A2}, \psi_{B1}\}$:

$$H^{AB} = \begin{pmatrix} \frac{\Delta}{2} & v_3\pi & 0 & v\pi^\dagger \\ v_3\pi^\dagger & -\frac{\Delta}{2} & v\pi & 0 \\ 0 & v\pi^\dagger & -\frac{\Delta}{2} & \gamma_1 \\ v\pi & 0 & \gamma_1 & \frac{\Delta}{2} \end{pmatrix} \tag{37}$$

The energy eigenvalue equation can be written as

$$\begin{pmatrix} h_\theta & u \\ u^\dagger & h_\chi \end{pmatrix} \begin{pmatrix} \theta \\ \chi \end{pmatrix} = E \begin{pmatrix} \theta \\ \chi \end{pmatrix}, \quad (38)$$

where

$$h_\theta = \begin{pmatrix} \frac{\Delta}{2} & v_3\pi \\ v_3\pi^\dagger & -\frac{\Delta}{2} \end{pmatrix}, \quad h_\chi = \begin{pmatrix} -\frac{\Delta}{2} & \gamma_1 \\ \gamma_1 & \frac{\Delta}{2} \end{pmatrix}, \quad u = u^\dagger = \begin{pmatrix} 0 & v\pi^\dagger \\ v\pi & 0 \end{pmatrix}. \quad (39)$$

We write the effective Hamiltonian such that $H_{\text{eff}}^{AB}\theta = E\theta$.

$$H_{\text{eff}}^{AB} \approx S^{-1/2} [h_\theta - uh_\chi^{-1}u^\dagger] S^{-1/2}, \quad S = 1 + uh_\chi^{-2}u^\dagger. \quad (40)$$

We assume $\gamma_0, \gamma_1 \ll |E|, vp, \gamma_3, \Delta$ so that we can expand the equation above and neglect higher order terms

$$H_{\text{eff}}^{AB} \approx \left(1 - \frac{1}{2}uh_\chi^{-2}u^\dagger\right) [h_\theta - uh_\chi^{-1}u^\dagger] \left(1 - \frac{1}{2}uh_\chi^{-2}u^\dagger\right) \approx h_\theta - uh_\chi^{-1}u^\dagger \quad (41)$$

We note that

$$h_\chi^{-1} = -\frac{1}{\Delta^2/4 + \gamma_1^2} \begin{pmatrix} \frac{\Delta}{2} & -\gamma_1 \\ -\gamma_1 & -\frac{\Delta}{2} \end{pmatrix}, \quad uh_\chi^{-2}u^\dagger = \frac{1}{\Delta^2/4 + \gamma_1^2} \begin{pmatrix} v^2\pi^\dagger\pi & 0 \\ 0 & v^2\pi\pi^\dagger \end{pmatrix} \quad (42)$$

$$\begin{aligned} uh_\chi^{-1}u^\dagger &= -\frac{1}{\frac{\Delta^2}{4} + \gamma_1^2} \begin{pmatrix} -\frac{\Delta}{2}v^2\pi^\dagger\pi & -\gamma_1v^2(\pi^\dagger)^2 \\ -\gamma_1v^2(\pi)^2 & \frac{\Delta}{2}v^2\pi^\dagger\pi \end{pmatrix} = \frac{1}{\gamma_1^2 \left(1 + \frac{\Delta^2}{4\gamma_1^2}\right)} \begin{pmatrix} -\frac{\Delta}{2}v^2\pi^\dagger\pi & -\gamma_1v^2(\pi^\dagger)^2 \\ -\gamma_1v^2(\pi)^2 & \frac{\Delta}{2}v^2\pi^\dagger\pi \end{pmatrix} \\ &\approx \begin{pmatrix} -\frac{\Delta}{2\gamma_1^2}v^2\pi^\dagger\pi & -\frac{v^2}{\gamma_1}(\pi^\dagger)^2 \\ -\frac{v^2}{\gamma_1}(\pi)^2 & \frac{\Delta}{2\gamma_1^2}v^2\pi^\dagger\pi \end{pmatrix} \approx \begin{pmatrix} 0 & -\frac{v^2}{\gamma_1}(\pi^\dagger)^2 \\ -\frac{v^2}{\gamma_1}(\pi)^2 & 0 \end{pmatrix} \end{aligned} \quad (43)$$

Thus the effective two-band Hamiltonian for AB stacking is

$$H_{\text{eff}}^{AB} \approx \begin{pmatrix} \frac{\Delta}{2} & v_3\pi \\ v_3\pi^\dagger & -\frac{\Delta}{2} \end{pmatrix} - \frac{1}{2m} \begin{pmatrix} 0 & (\pi^\dagger)^2 \\ (\pi)^2 & 0 \end{pmatrix}, \quad (44)$$

where $m = v^2/\gamma_1$.

3.2 BA stacking

For BA stacking, the non-dimer site consists of $\theta = \{\psi_{A2}, \psi_{B1}\}$, and the dimer site consists of $\chi = \{\psi_{A1}, \psi_{B2}\}$. We rewrite H^{BA} in the basis of $\{\psi_{A2}, \psi_{B1}, \psi_{A1}, \psi_{B2}\}$:

$$H^{BA} = \begin{pmatrix} -\frac{\Delta}{2} & v_3\pi & 0 & v\pi^\dagger \\ v_3\pi^\dagger & \frac{\Delta}{2} & v\pi & 0 \\ 0 & v\pi^\dagger & \frac{\Delta}{2} & \gamma_1 \\ v\pi & 0 & \gamma_1 & -\frac{\Delta}{2} \end{pmatrix} \quad (45)$$

In this case, we have

$$h_\theta = \begin{pmatrix} -\frac{\Delta}{2} & v_3\pi \\ v_3\pi^\dagger & \frac{\Delta}{2} \end{pmatrix}, \quad h_\chi = \begin{pmatrix} \frac{\Delta}{2} & \gamma_1 \\ \gamma_1 & -\frac{\Delta}{2} \end{pmatrix}, \quad u = u^\dagger = \begin{pmatrix} 0 & v\pi^\dagger \\ v\pi & 0 \end{pmatrix}. \quad (46)$$

We note that the only difference from the AB stacking is that $\Delta \rightarrow -\Delta$. Thus BA stacking is equivalent to AB stacking with a reversed out-of-plane electric field.

4 Peierls Substitution

In the presence of an external vector potential, we write

$$H_{\alpha\beta} = \gamma_{\alpha\beta} f(\mathbf{k}) e^{iq \int_{\mathbf{R}_\alpha}^{\mathbf{R}_\beta} \mathbf{A} \cdot d\mathbf{r}} \quad (47)$$

We assume that the vector potential does not vary in the in-plane direction but exponentially decays in the z-direction $\mathbf{A} = \mathbf{A}_0 e^{-(z+z_0)/\lambda_z}$.

For intralayer coupling, $\mathbf{A}$ is constant when hopping between nearest neighbours and we have

$$H_{A1,B1} = \gamma_0 \sum_{l=1}^3 e^{iq \mathbf{A} \cdot \boldsymbol{\delta}_l + i\mathbf{k} \cdot \boldsymbol{\delta}_l} \quad (48)$$

For interlayer coupling, $\mathbf{A}$ decays exponentially between one layer to the other. We can separate the contributions from the in-plane field and the out-of-plane field A_z :

$$H_{A1,A2} = \gamma_4 f(\mathbf{k}) \sum_{l=1}^3 e^{iq \int_{\delta_l} \mathbf{A} \cdot d\mathbf{r}} = \gamma_4 f(\mathbf{k}) \sum_{l=1}^3 e^{iq \int_0^{a/\sqrt{3}} \mathbf{A} \cdot \hat{\boldsymbol{\delta}}_l ds + iq \int A_z dz} \quad (49)$$

$$= \gamma_4 f(\mathbf{k}) \sum_{l=1}^3 \exp \left[iq \int_0^{a/\sqrt{3}} \mathbf{A}_0 \cdot \hat{\boldsymbol{\delta}}_l e^{-(s\delta_z/a + z_0)/\lambda_z} ds + iq \int_0^{\delta_z} A_{z0} e^{-(z+z_0)/\lambda_z} dz \right] \quad (50)$$

$$= \gamma_4 f(\mathbf{k}) \sum_{l=1}^3 \exp [iq(\mathbf{A}_0 \cdot \boldsymbol{\delta}_l \phi_1 + A_{z0} \delta_z \phi_2)] \quad (51)$$

$$= \gamma_4 \sum_{l=1}^3 e^{i\mathbf{k} \cdot \boldsymbol{\delta}_l + iq\phi_1 \mathbf{A}_0 \cdot \boldsymbol{\delta}_l} e^{iq\phi_2 A_{z0} \delta_z} \quad (52)$$

$$H_{A1,B2} = \gamma_3 \sum_{l=1}^3 e^{-i\mathbf{k} \cdot \boldsymbol{\delta}_l - iq\phi_1 \mathbf{A}_0 \cdot \boldsymbol{\delta}_l} e^{iq\phi_2 A_{z0} \delta_z} \quad (53)$$

$$H_{B1,A2} = \gamma_1 e^{iq\phi_2 A_{z0} \delta_z} \quad (54)$$

By taking position of the bottom layer to be $z_0 = 2$ nm, interlayer separation $\delta_z = 0.3$ nm and confinement length $\lambda_z = 5$ nm, we have

$$\phi_1 = -\frac{\lambda_z}{\delta_z} \left(e^{-(\delta_z+z_0)/\lambda_z} - e^{-z_0/\lambda_z} \right) \approx 0.651 \quad (55)$$

$$\phi_2 = \frac{\lambda_z}{\delta_z} \left(e^{-z_0/\lambda_z} - e^{-(\delta_z+z_0)/\lambda_z} \right) = \phi_1 \approx 0.651 \quad (56)$$

The field strength at the bottom layer (layer 1) is $\mathbf{A}(z_0) = \mathbf{A}_0 e^{-z_0/\lambda} \approx 0.670 \mathbf{A}_0$ and the field at the top layer is $\mathbf{A}(z_0 + \delta_z) = \mathbf{A}_0 e^{-(z_0+\delta_z)/\lambda_z} \approx 0.631 \mathbf{A}_0$.